



A method for the calibration of a 3-D laser scanner[☆]

Vincenzo Niola^{*}, Cesare Rossi¹, Sergio Savino, Salvatore Strano

Department of Mechanics and Energetics (DiME), University of Naples "Federico II", via Claudio 21, Naples 80125, Italy

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ABSTRACT

The calibration of a three-dimensional digitizer is a very important issue to take into consideration that good quality, reliability, accuracy and high repeatability are the features which a good digitizer is expected to have. The aim of this paper is to propose a new method for the calibration of a 3-D laser scanner, mainly for robotic applications. The acquisition system consists of a laser emitter and a webcam with fixed relative positions. In addition, a cylindrical lens is provided with the laser housing so that it is capable to project a plane light. An optical filter was also used in order to segment the laser stripe from the rest of the scene. For the calibration procedure it was used a digital micrometer that move a target with known dimensions. The calibration method is based on the modeling of the geometrical relationship between the 3-D coordinates of the laser stripe on the target and its digital coordinates in the image plane. By this method it is possible to calibrate the intrinsic parameters of the video system, the position of the image plane and the laser plane in a given frame, all in the same time.

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1. Introduction

Calibration is a basic requirement in multi-sensor platforms where data need to be represented in a common reference frame for the purpose of analysis and data fusion. On platforms where a camera provides intensity information in the form of an image and a laser supplies depth information in the form of a set of 3-D points, external calibration allows reprojection of the 3-D points from the laser coordinate frame to the 2-D coordinate frame of the image. The process of external calibration is often poorly documented and is almost always notoriously laborious. It usually involves some modification of the scene in the form of markers with features visible by both the camera and the laser. These features are often in the form of edges or corners extracted from the laser data with considerable user intervention. In procedures where the features are edges or corners, the quality of the final estimates is dependent on the accuracy by which features can be localized and so requires the ability to obtain laser data at very high resolution.

The procedure proposed in this document uses nothing more than the same checkerboard calibration target commonly used for internal calibration of the camera. An interactive GUI allows the user to acquire the images of laser on a target that can move along camera optical axis, by means of a digital micrometer. A robust fitting procedure fits geometrical relationship between the 3-D

coordinates of the laser stripe on the target and its digital coordinates in the image plane, based on triangulation principle.

By this method it is possible to calibrate the intrinsic parameters of the video system, the position of the image plane and the laser plane in a given frame, all in the same time.

2. 3-D reconstruction: principle of operation

There are many methods in order to reconstruct the object shape by means of videos or images [1]. In this paper is described a system, that consists of a linear laser emitter and a webcam, and uses triangulation principle applied to a scanning belt on object surface [2].

Camera observes the intersection between laser and object: laser line points in image frame are the intersections between image plane and optical rays that pass through the intersection points between laser and object. By means of a transformation matrix, it is possible to express the image frame coordinates, in pixel, in a local reference frame. In Fig. 1(a) is shown a scheme of scanning system: $\{W\}$ is local reference frame, $\{I\}$ is image frame with coordinates system $\{u, v\}$ and $\{L\}$ is laser frame. $\{L2\}$ is laser plane that contains laser knife and scanning belt on the surface of object, and it coincides with (x, y) plane of laser frame $\{L\}$, [3].

Starting from the coordinates in pixel (u, v) , in image frame, it is possible to write the coordinates of the scanning belt on the object surface, in camera frame by means of Eq. (1). Camera frame is located in camera focal point Fig. 1(b)

$$\begin{pmatrix} x_c \\ y_c \\ z_c \\ 1 \end{pmatrix} = \begin{bmatrix} \delta_x & 0 & 0 & -\delta_x u_0 \\ 0 & \delta_y & 0 & -\delta_y v_0 \\ 0 & 0 & 0 & f \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} u \\ v \\ 0 \\ 1 \end{pmatrix} \quad (1)$$

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^{*} Corresponding author. Tel.: +39 081 7683482; fax: +39 081 2394165.

E-mail addresses: vincenzo.niola@unina.it (V. Niola), cesare.rossi@unina.it (C. Rossi).

¹ Tel.: +39 081 7683269; fax: +39 081 2394165.

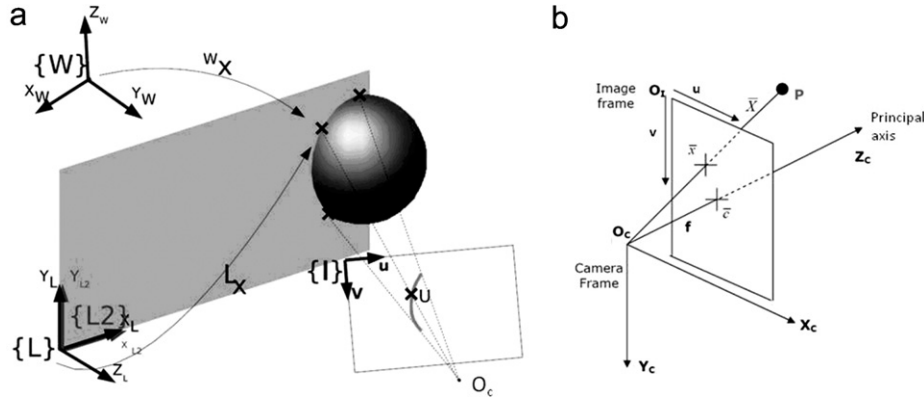


Fig. 1. Reference frames.

with (u_0, v_0) is the image frame coordinates of focal point projection in image plane; (δ_x, δ_y) the physical dimension of sensor pixel along directions u and v ; f the focal length.

It is possible to write the expression of the optical beam of a generic point in the image frame, that can be identified by means of parameter t .

$$\begin{cases} x_c = (u - u_0)\delta_x t \\ y_c = (v - v_0)\delta_y t \\ z_c = ft \end{cases} \quad (2)$$

Laser frame $\{L\}$ is rotated and translated with respect to camera frame, the steps and their sequence are:

- I. Translation Δx_{lc} along axis x_c .
- II. Translation Δy_{lc} along axis y_c .
- III. Translation Δz_{lc} along axis z_c .
- IV. Rotation ϕ_{lc} around axis z_c .
- V. Rotation θ_{lc} around axis y_c .
- VI. Rotation ψ_{lc} around axis x_c .

Hence in Eq. (3) the transformation matrix between laser frame and camera frame is

$${}^cT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\psi_{lc}) & -\sin(\psi_{lc}) & 0 \\ 0 & \sin(\psi_{lc}) & \cos(\psi_{lc}) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(\theta_{lc}) & 0 & \sin(\theta_{lc}) & 0 \\ 0 & 1 & 0 & 0 \\ -\sin(\theta_{lc}) & 0 & \cos(\theta_{lc}) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ \times \begin{bmatrix} \cos(\phi_{lc}) & -\sin(\phi_{lc}) & 0 & 0 \\ \sin(\phi_{lc}) & \cos(\phi_{lc}) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \Delta z_{lc} \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ \times \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & \Delta y_{lc} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & \Delta x_{lc} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (3)$$

Laser plane $\{L2\}$ coincides with (x, y) plane of laser frame $\{L\}$, so it contains three points

$$\{p\}_l = \{0, 0, 0, 1\}^T; \quad \{q\}_l = \{1, 0, 0, 1\}^T; \quad \{r\}_l = \{0, 0, 1, 1\}^T \quad (4)$$

In camera frame

$$\begin{aligned} \{p_x, p_y, p_z, 1\}_c^T &= {}^cT^{-1} \{p\}_l; \quad \{q_x, q_y, q_z, 1\}_c^T \\ &= {}^cT^{-1} \{q\}_l; \quad \{r_x, r_y, r_z, 1\}_c^T = {}^cT^{-1} \{r\}_l \end{aligned} \quad (5)$$

It is possible to obtain laser plane equation in camera frame, solving Eq. (6) in x_c, y_c and z_c .

$$\det \begin{bmatrix} x_c - p_x & y_c - p_y & z_c - p_z \\ q_x - p_x & q_y - p_y & q_z - p_z \\ r_x - p_x & r_y - p_y & r_z - p_z \end{bmatrix} = 0 \quad (6)$$

If it is

$$\begin{aligned} M_x &= \det \begin{bmatrix} q_y - p_y & q_z - p_z \\ r_y - p_y & r_z - p_z \end{bmatrix}; \quad M_y = \det \begin{bmatrix} q_x - p_x & q_z - p_z \\ r_x - p_x & r_z - p_z \end{bmatrix}; \\ M_z &= \det \begin{bmatrix} q_x - p_x & q_y - p_y \\ r_x - p_x & r_y - p_y \end{bmatrix} \end{aligned}$$

$\{L2\}$ equation in camera frame is

$$(x_c - p_x)M_x - (y_c - p_y)M_y + (z_c - p_z)M_z = 0 \quad (7)$$

It is possible to evaluate coordinates x_c, y_c and z_c in camera frame, solving system (8) with unknown t

$$\begin{cases} x_c = (u - u_0)\delta_x t \\ y_c = (v - v_0)\delta_y t \\ z_c = ft \\ (x_c - p_x)M_x - (y_c - p_y)M_y + (z_c - p_z)M_z = 0 \end{cases} \quad (8)$$

The solution is

$$t = \frac{p_x M_x - p_y M_y + p_z M_z}{(u - u_0)\delta_x M_x - (v - v_0)\delta_y M_y + f M_z} \quad (9)$$

Eq. (9) permits to compute in the camera frame, the points coordinates of the scanning belt on the object surfaces, starting to its image coordinates (u, v) . In this way it is possible to do a 3-D objects reconstruction by means of a laser knife.

3. Detection of the laser path on the target

A very important step for 3-D reconstruction is image elaboration for the laser path on the target [4–6,8,9], the latter is shown in Fig. 2.

Image elaboration procedure permits the user to choose some image points of laser line, in order to identify three principal colours (red, green and blue) of laser line, Fig. 3.

With mean values of scanning belt principal colours, it is possible to define a brightness coefficient of the laser line, according to relation (10)

$$s = \frac{\max(\text{mean}(R), \text{mean}(G), \text{mean}(B)) + \min(\text{mean}(R), \text{mean}(G), \text{mean}(B))}{2} \quad (10)$$

By means of relation (11), an intensity analysis is carried out on RGB image.

$$L(u, v) = \frac{\max(R, G, B) + \min(R, G, B)}{2} \quad (11)$$

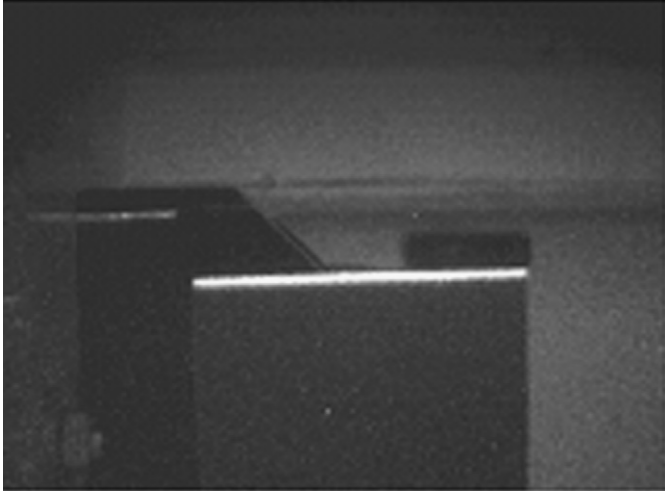


Fig. 2. Laser path image.

By Eq. (11), the matrix that contains the three layers of RGB image, is transformed in a matrix L , that represents image intensity. This matrix represents same initial image, but it gives information only about luminous intensity in each image pixel, and so it is a grayscale expression of initial RGB image; examples of the above is shown in Fig. 4(a) and (b).

With relation (12), it is possible to define a logical matrix I_b . Matrix I_b indicates pixels of matrix L with a brightness in a range of 15% brightness coefficient s .

$$I_b(u, v) = \begin{cases} 1 & \text{for } 0.85s \leq L(u, v) \leq 1.15s \\ 0 & \text{otherwise} \end{cases} \quad (12)$$

In Fig. 4(c), matrix I_b is shown.

In matrix I_b , the scanning belt on the object surface is represented by means of the pixel value one. The area of the laser path in the image plane depends on the real dimension of laser beam and on external factors such as: reflection phenomena, inclination of object surfaces. In Fig. 5 an example is reported.

3-D reconstruction procedure is based on triangulation principle and it does not consider the laser beam thickness, so it is necessary to associate a line to the image of scanning belt.

Since the laser path in the image plane is rather horizontal, a geometrical mean is computed on the columns of the matrix I_b , that is to say: in the opposite direction to wider extension of the laser path in the image. This is shown in Eq. (13).

$$\Omega = \{u, v\}; \quad \bar{\Omega} = \{\bar{u}, \bar{v}\}$$

$$\Rightarrow \left\{ \bar{u} = \frac{\sum_{u, v \in \Omega} u I_b(u, v)}{\sum_{u, v \in \Omega} I_b(u, v)} \neq \infty, \bar{v} = v : \sum_{u, v \in \Omega} I_b(u, v) \neq 0 \right\} \quad (13)$$

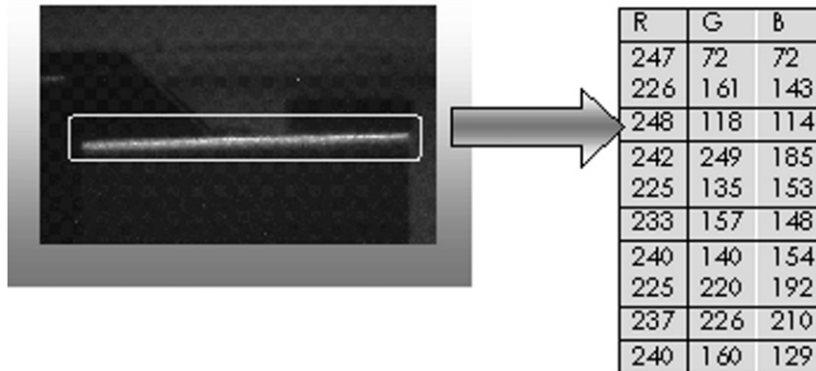
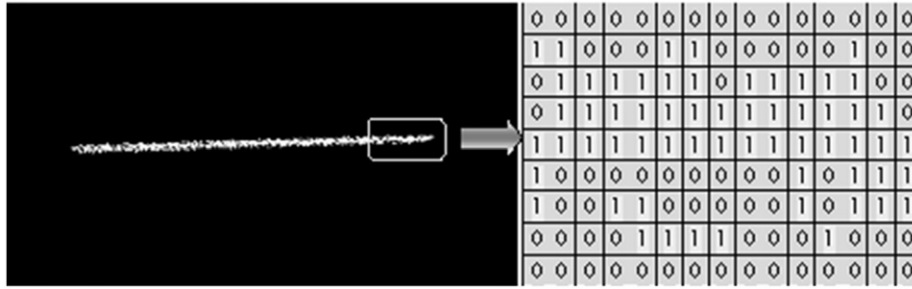
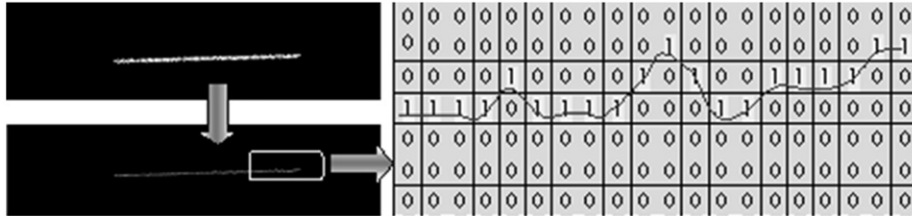
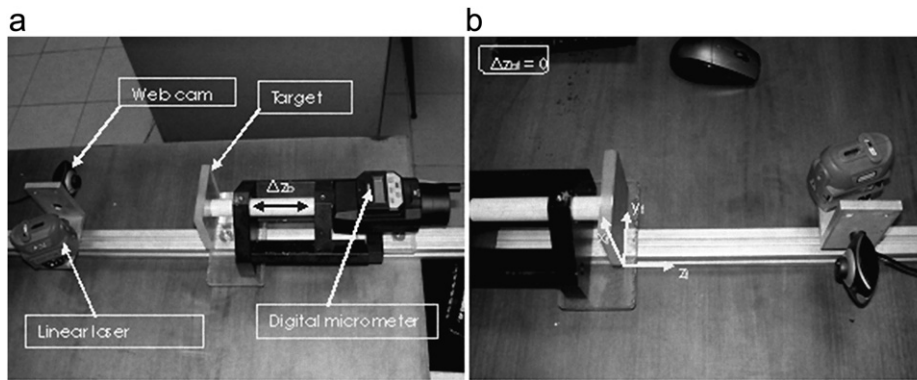


Fig. 3. Laser line RGB representation.



Fig. 4. (a) RGB initial image; (b) grayscale initial image L ; matrix I_b .

Fig. 5. I_b representation.Fig. 6. h_b representation.Fig. 7. (a) Laser scanner module calibration test rig; (b) fixed frame $O_{x_f y_f z_f}$.

Then a matrix h_p is also defined as follows

$$h_b(u,v) = \begin{cases} 1 & \text{se } \{u,v\} \in \overline{\Omega} \\ 0 & \text{otherwise} \end{cases} \quad (14)$$

Matrix h_b is hence a logical matrix with same dimension of I_b in which that represents laser image like a line, see Fig. 6. This line is centre line of scanning belt on object surface in image.

The set of transformations (11)–(13) represents the image elaboration, that is necessary to identify a laser line in image. The points of this line are used in 3-D reconstruction procedure.

4. Calibration system

It is necessary to identify all model parameters in order to obtain a good 3-D reconstruction. Calibration is a basic procedure for the data analysis [3,5,7,10–12]. For this reason, a calibration procedure for a classic laser scanner module was developed.

The calibration procedure of the laser scanner system is to determine the relative positions between the laser module and camera. The relationship between laser module and camera is achieved through a procedure implemented by maintaining the fixed position of the scanner module in front of a moving target; see

Figs. 7 and 8. One set of images is acquired for different positions of the target. At each step it is possible to evaluate the coordinates, in the image plane, of the points of the laser line projected on the target. By means of an optimization procedure all the parameters of the transformation that describe the relationship between laser and camera can be identified.

The laser scanner module is composed by a webcam with resolution 640×480 pixel and a linear laser. The calibration test rig is realized with a guide on which is fixed the laser scanner module and a digital micrometer with a target, Fig. 7(a).

A fixed frame $O_{x_f y_f z_f}$ has origin in a vertex of the rectangular target with dimensions $a = 68.0$ mm and $b = 74.5$ mm, like it is shown in Fig. 7(b). Target movements are indicated with Δz_{bf} .

In Fig. 8 some steps of calibration procedure are shown.

Calibration procedure is based on a set of images, taken with the target placed in different positions respect to the laser scanner module. A least square optimization allows to identify all parameters. In Fig. 9 are shown 20 images used for calibration, of scanning belt, with 20 different Δz_{bf} .

The aim of the calibration procedure is to obtain all the parameters that define the relation between the laser frame $Lx_L y_L z_L$ and the camera frame $O_c x_c y_c z_c$; see Fig. 1. In particular, it is possible to calibrate the intrinsic parameters of the video system, the

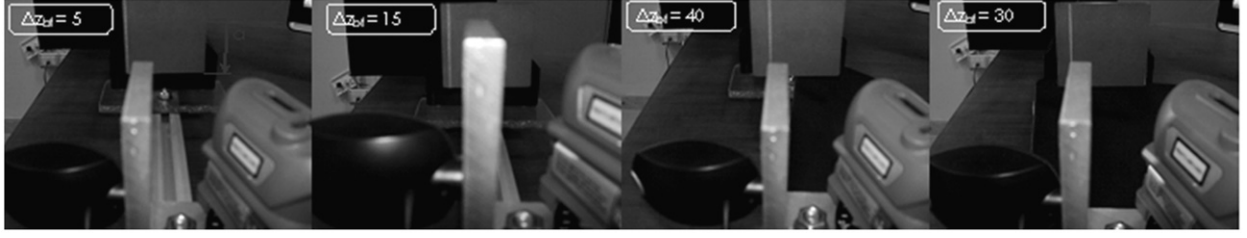


Fig. 8. Calibration procedure steps.

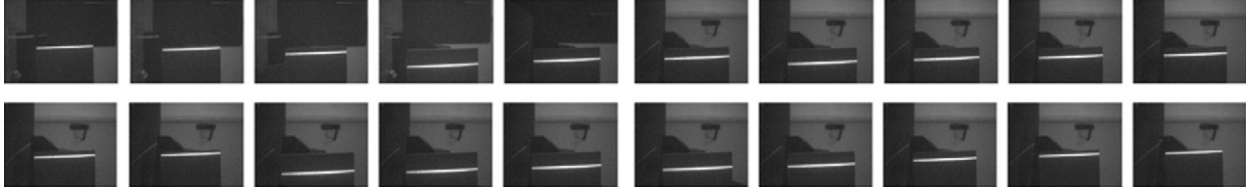


Fig. 9. Example of a set of 20 images used for calibration.

position of the image plane and the laser plane in a given frame, all in the same time.

In every calibration step, it is possible to define transformation matrix between fixed frame $O_{\alpha\beta\gamma}z_f$ and target frame

$$[{}^fT]_i = \begin{bmatrix} 1 & 0 & 0 & 0; & 0 & 1 & 0 & 0; & 0 & 0 & 1 & -\Delta z_{fb}^i; & 0 & 0 & 0 & 1 \end{bmatrix} \quad (15)$$

The transformation matrix between fixed frame $O_{\alpha\beta\gamma}z_f$ and camera frame $O_{c\alpha_c\beta_c\gamma_c}z_c$ can be obtained analogous to matrix $[{}^cT]_i$, and it is a function of six parameters

$$[{}^fT] = f(\Delta x_{cf}, \Delta y_{cf}, \Delta z_{cf}, \psi_{cf}, \theta_{cf}, \phi_{cf}) = f(\pi_{cf}) \quad (16)$$

with π_{cf} set of parameter of transformation $[{}^fT]_c$.

Eq. (3) can be written as

$$[{}^cT] = g(\Delta x_{lc}, \Delta y_{lc}, \Delta z_{lc}, \psi_{lc}, \theta_{lc}, \phi_{lc}) = f(\pi_{lc}) \quad (17)$$

with π_{lc} set of parameter of transformation $[{}^cT]_l$.

For each step of the calibration procedure it is possible to evaluate the coordinates of the points of the laser line in the camera frame, according to Eqs. (5) and (13)

$$\{\bar{x}_c, \bar{y}_c, \bar{z}_c, 1\}_i^T = f(\pi_{lc}, f) \cdot \{\bar{u}, \bar{v}, 0, 1\}_i^T \quad (18)$$

By means of Eq. (16), it is possible to write

$$\begin{aligned} \{\bar{x}_f, \bar{y}_f, \bar{z}_f, 1\}_i^T &= [{}^fT]^{-1} \{\bar{x}_c, \bar{y}_c, \bar{z}_c, 1\}_i^T = [{}^fT]^{-1} f(\pi_{lc}, f) \{\bar{u}, \bar{v}, 0, 1\}_i^T \\ &= \omega(\pi_{cf}, \pi_{lc}, f) \{\bar{u}, \bar{v}, 0, 1\}_i^T \end{aligned} \quad (19)$$

The optimization problem can be defined as

$$\min_{\rho \in R^{13}} F(\rho)^2 \quad (20)$$

$F(\rho)$ is a vectorial function defined as

$$F(\rho) = \{(\bar{z}_f^{i+1} - \bar{z}_f^i) - (\Delta \bar{z}_{bf}^{i+1} - \Delta \bar{z}_{bf}^i), \min(\bar{x}_f^i), \max(\bar{x}_f^i) - a, \max(\bar{z}_f^i) - \min(\bar{z}_f^i), \max(\bar{y}_f^i) - \min(\bar{y}_f^i), \min(\bar{y}_f^1), \min(\bar{z}_f^1)\}^T \quad (21)$$

With a least square optimization, the unknown parameters of set $[\pi_{cf}, \pi_{lc}, f]$ are identified.

An interactive GUI was developed to allow the user to acquire images, to execute optimization and to verify the results.

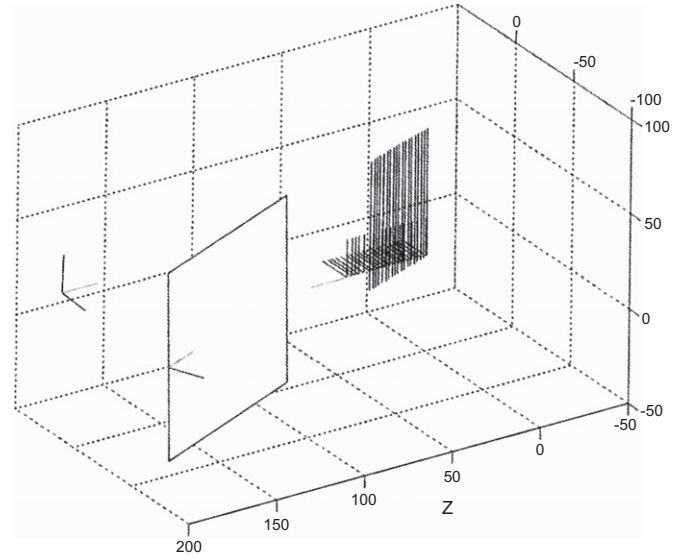


Fig. 10. Calibration results.

In Fig. 10, a graphical result of calibration is shown: it is possible to observe the camera frame position, the laser beam (represented by the quadrilateral on the left) and the 3-D reconstructions of scanning beam on target surface, for each image used in the calibration procedure.

5. Conclusions

A method for calibration of a 3-D laser scanner has been presented. The procedure is absolutely non-invasive since it does not involve any modification of the scene; in fact no markers with features visible by both the camera and the laser, or any other device, are required.

The method essentially uses a laser emitter and a webcam with fixed relative positions. A cylindrical lens and an optical filter are also provided; the first one is in order to project a plane light, while the second one was used to segment the laser stripe from the rest of the scene. The calibration method is based on the modeling of the geometrical relationship between the 3-D coordinates of the laser stripe on the target and its digital coordinates in the image plane.

By this method it is possible to calibrate the intrinsic parameters of the video system, the position of the image plane and the laser plane in a given frame, all in the same time.

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